

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_12apr25\_isolado.parquet

model\_type: LSTM  
group\_sec: 80  
timesteps\_orig: 60  
batch\_size: 256  
min\_units: 16  
max\_units: 256  
units\_step: 16  
dropout\_min: 0.2  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 20  
hyperband\_iterations: 3  
group\_msec: 80  
samples\_per\_sec: 12  
forecast\_horizon\_sec: 4  
forecast\_steps: 48  
timesteps: 60

### LSTM Multi-Step Training Report

Report Generated: 2025-08-01 13:56:06

Training Start: 2025-08-01 13-39-57

Training End: 2025-08-01 13-45-03

Training Duration: 5.11 minutes

Predictions Start: 2025-08-01 13-45-03

Predictions End: 2025-08-01 13-45-40

Predictions Duration: 0.62 minutes

#### Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 2.6614

MSE: 9.9148

RMSE: 3.1488

R2 Score: -0.3177

#### Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0716

MSE: 0.0074

RMSE: 0.0859

R2 Score: 0.4648

Dataset size before down-sample: 31,097 rows  
Dataset size after down-sample: 4,672 rows

First 5 timestamps BEFORE:

1744464597710  
1744464597720  
1744464597744  
1744464597757  
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

First 5 timestamps AFTER:

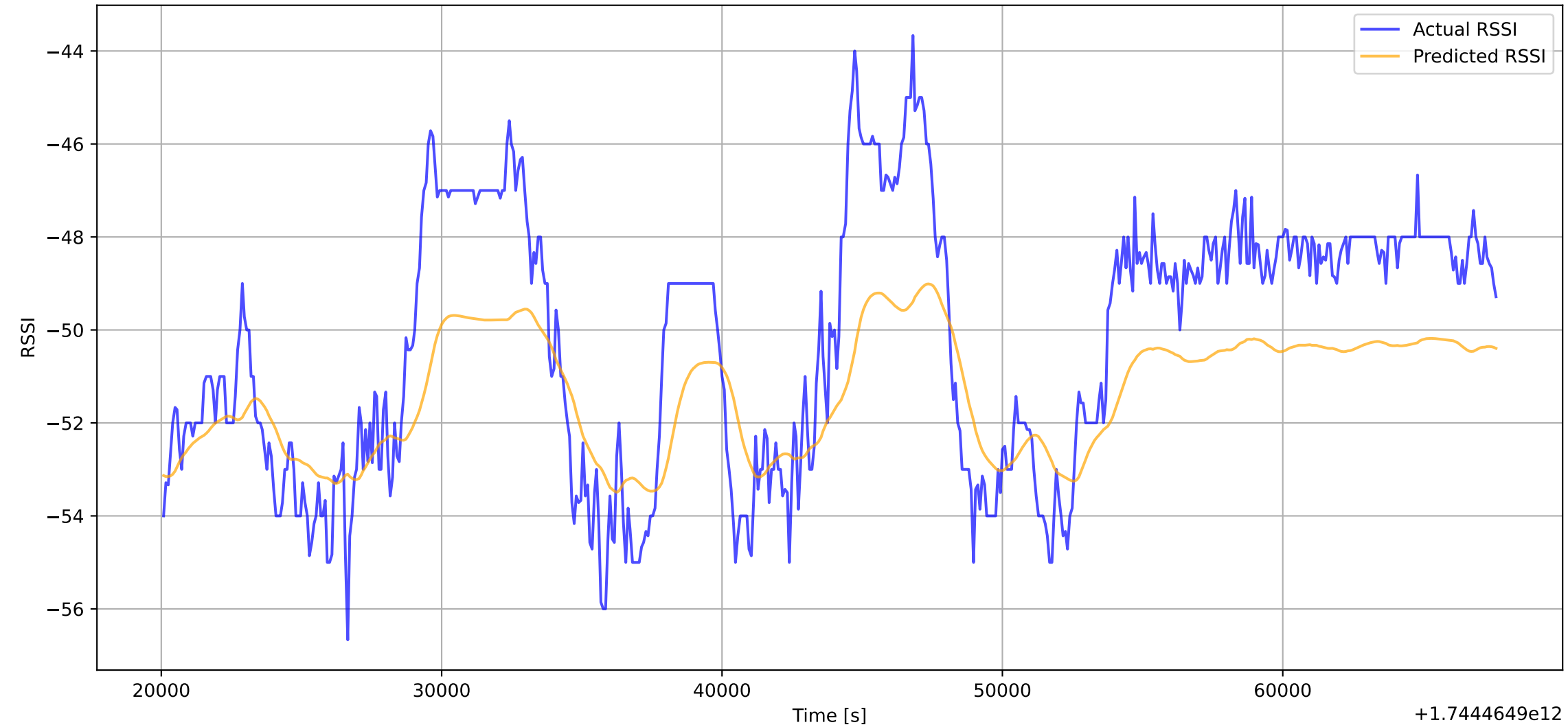
1744464597710  
1744464597783  
1744464597879  
1744464597921  
1744464598008

...

Last 5 timestamps AFTER:

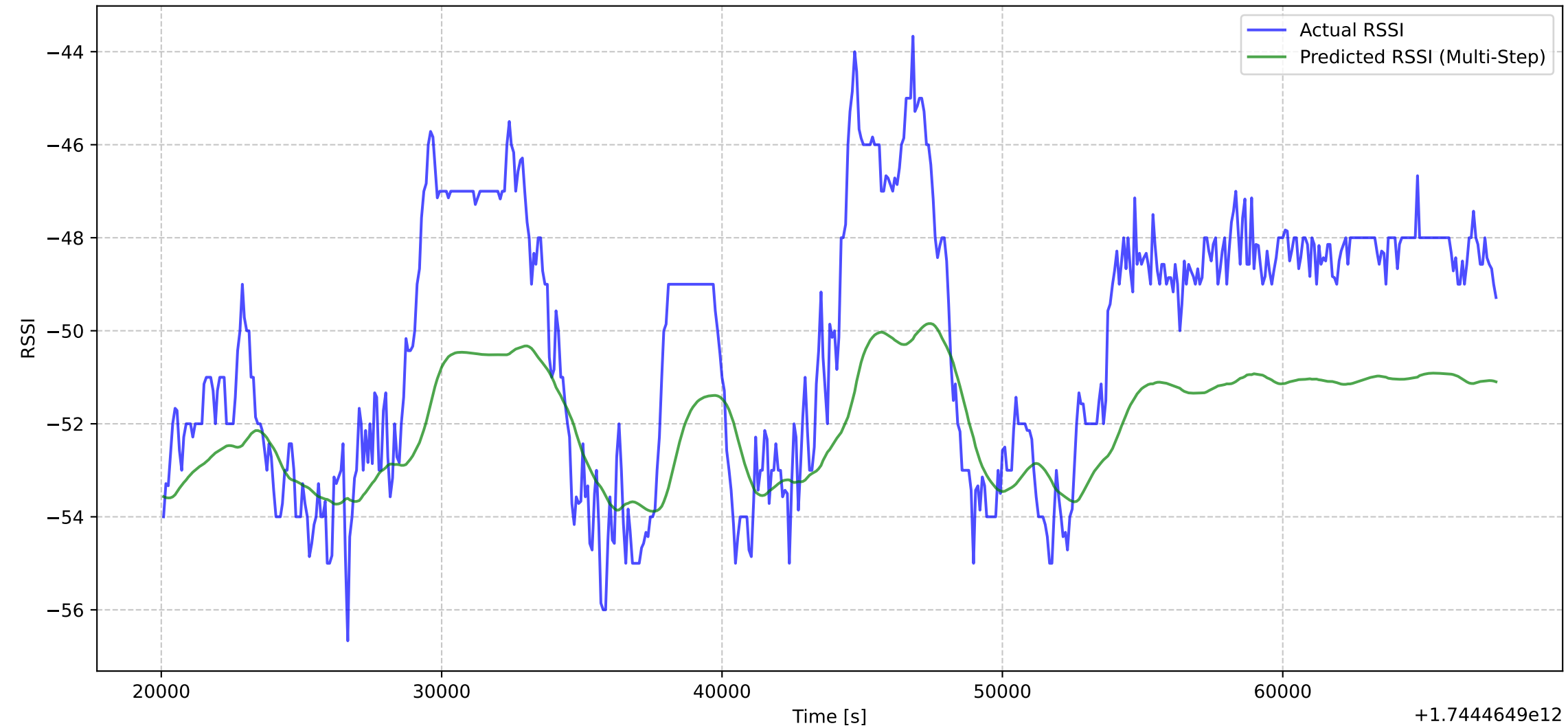
1744464971041  
1744464971124  
1744464971209  
1744464971281  
1744464971366

Actual vs Predicted RSSI (First Step)

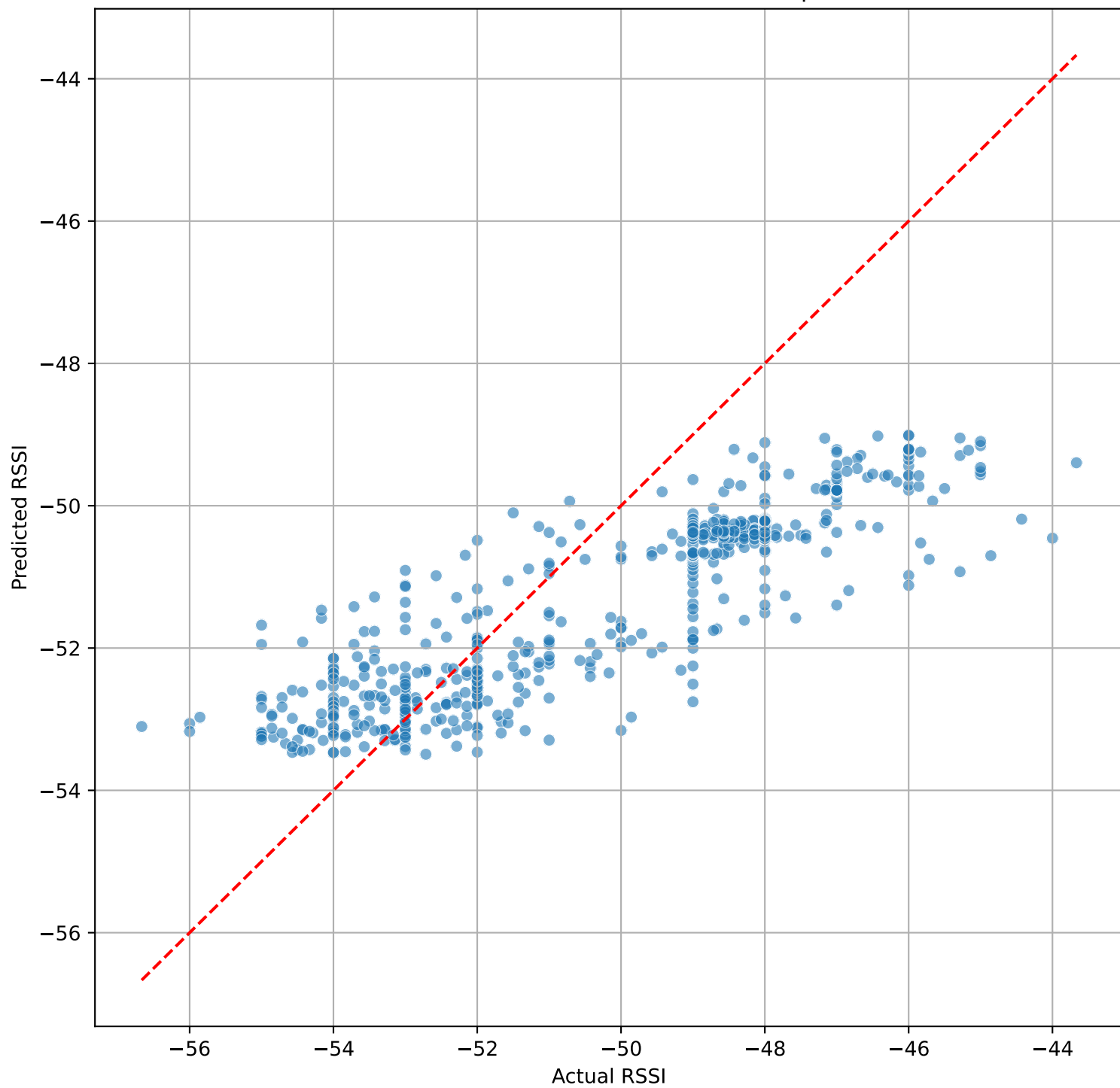


+1.7444649e12

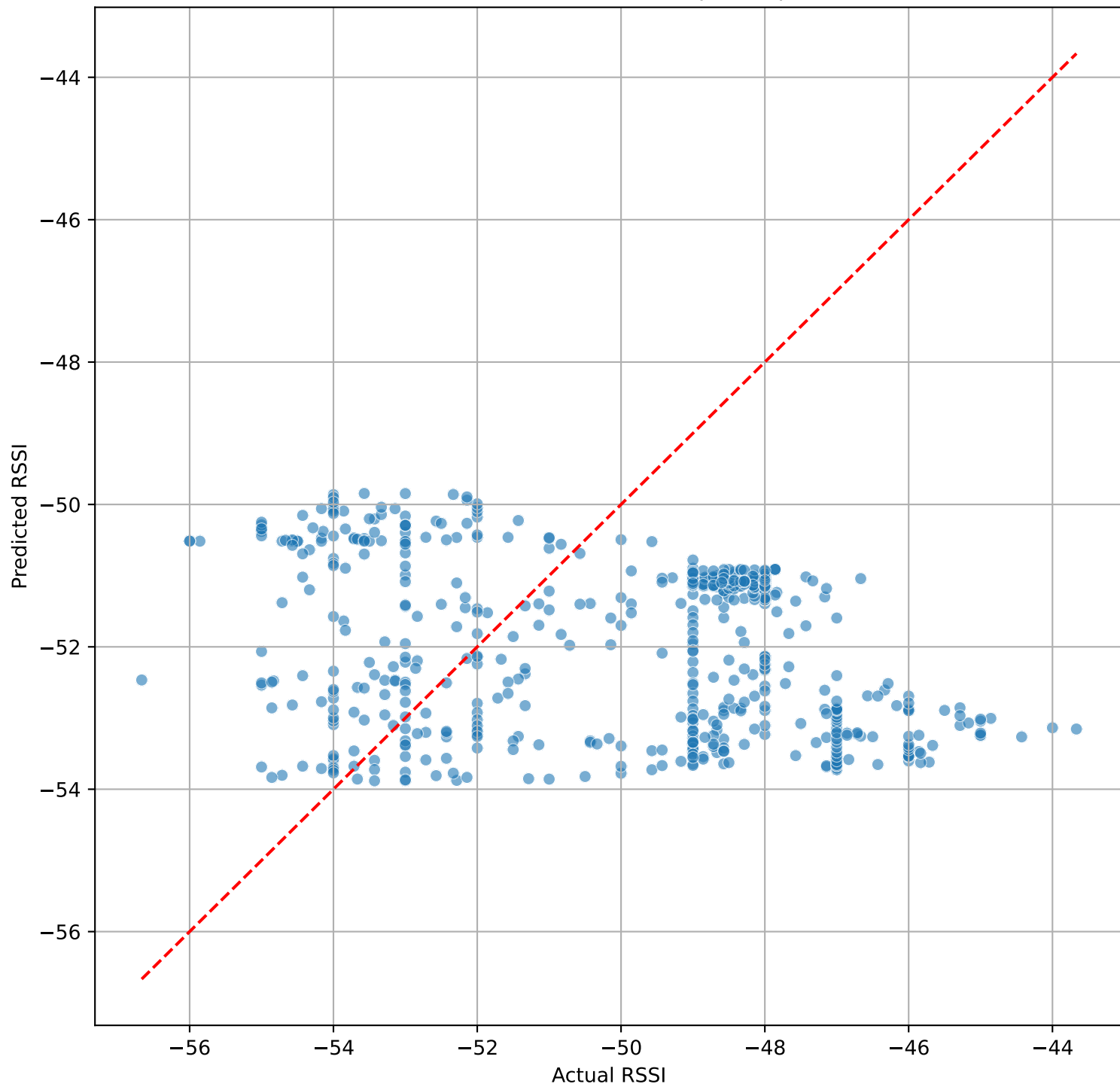
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$  4.0 s))



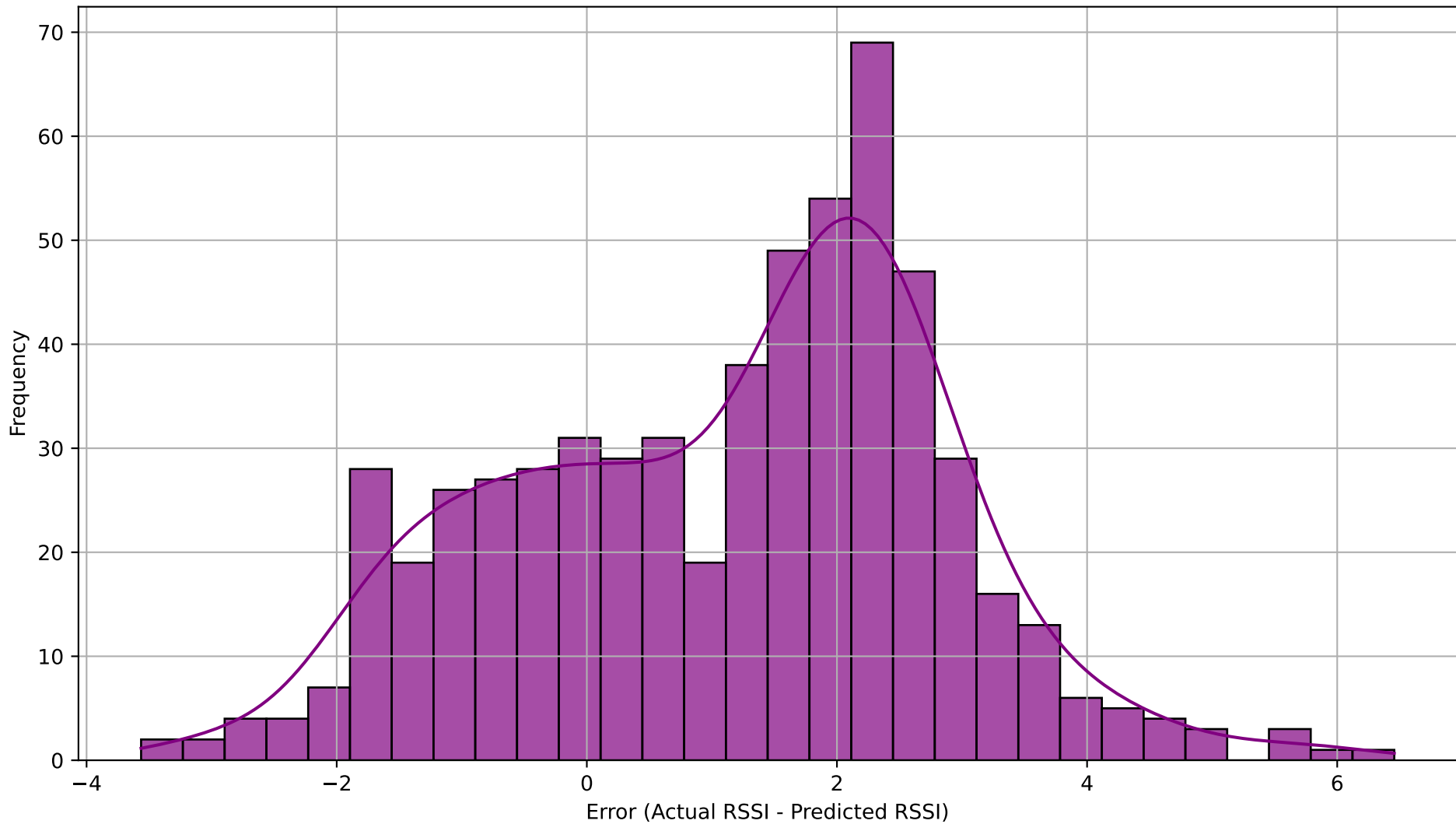
Actual vs Predicted RSSI (First Step)



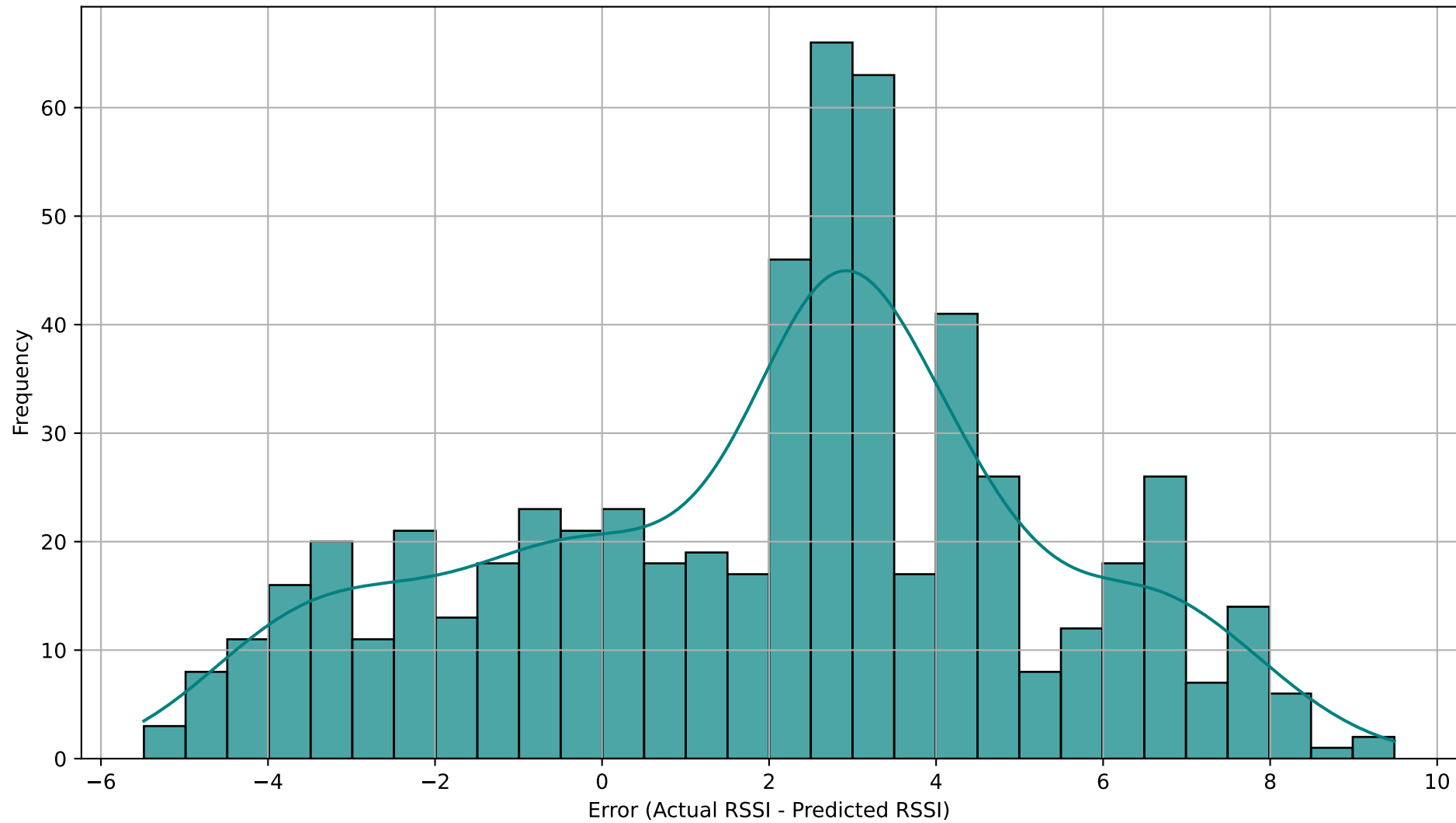
Actual vs Predicted RSSI (Last Step - step 48  $\approx$  4.0 s)



Distribution of Prediction Errors (First Step)

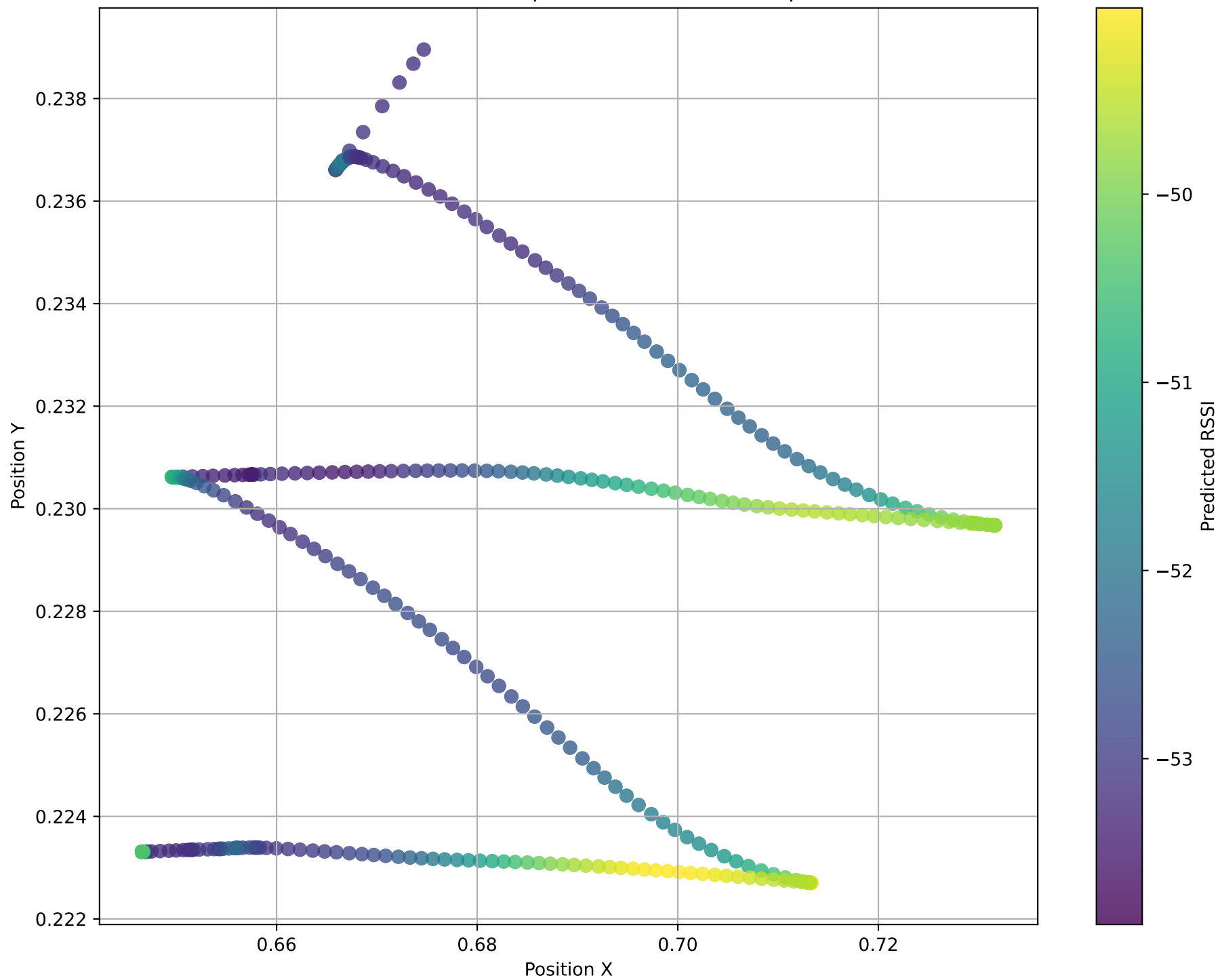


Distribution of Prediction Errors (Last Step - step 48  $\approx$  4.0 s)

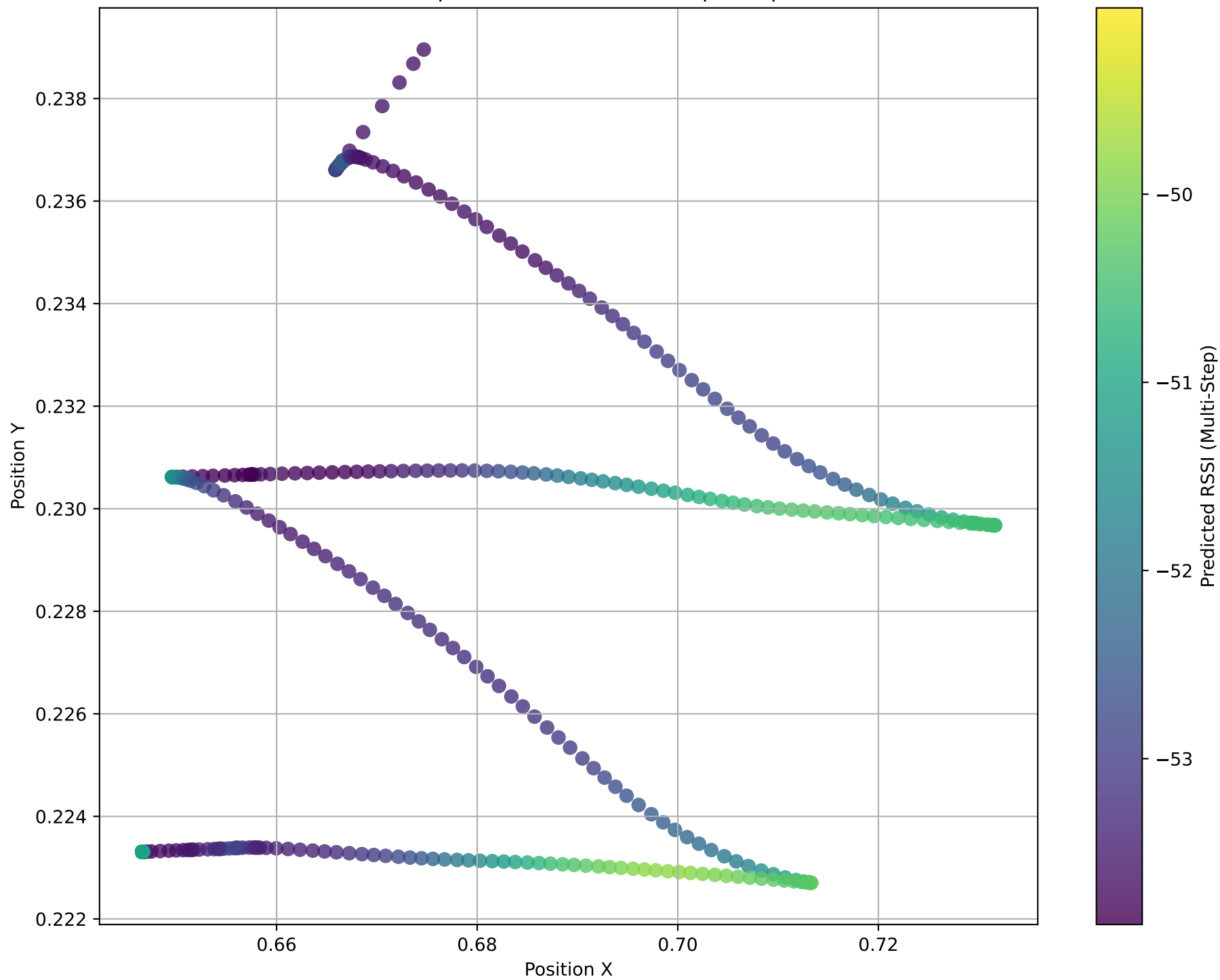




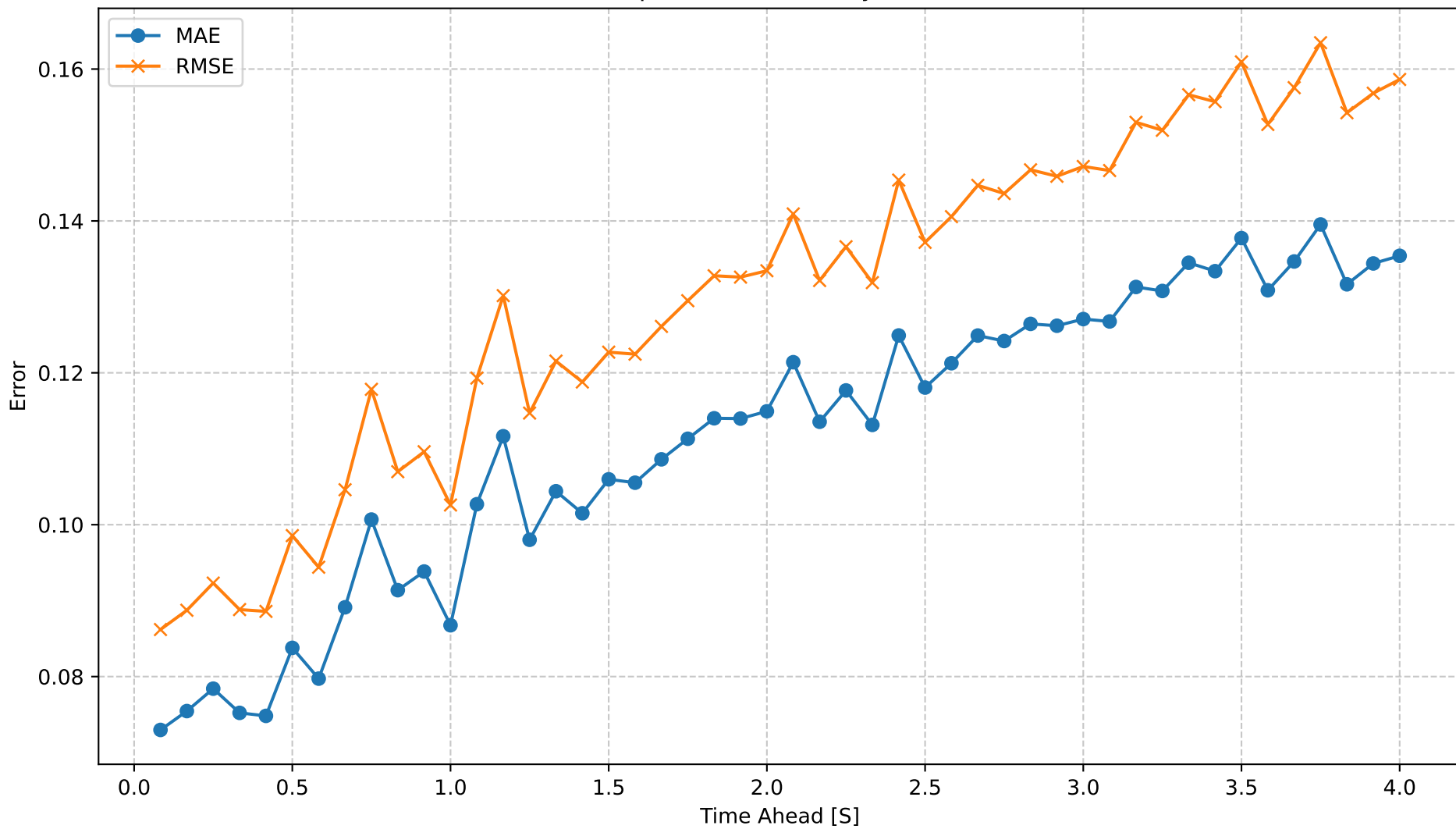
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48  $\approx$  4.0 s)



Multi-Step Forecast Error by Time Ahead



### Resumo Automático

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Dataset: rssi\_odom\_dataset\_12apr25\_isolado.parquet  
Modelo: LSTM  
Forecast steps: 48

#### MAE / RMSE (dB)

- 1º passo : 0.07 / 0.09
- Mediana : 0.11 / 0.13
- 48º passo : 0.14 / 0.16

Crescimento do erro: 85.6% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 45 (0.14), 42 (0.14), 48 (0.14)

#### Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.